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Printed Page: 1 of 1

Subject Code: BP401T

Roll No:

BPHARM

(SEM. IV) THEORY EXAMINATION 2025-26

PHARMACEUTICAL ORGANIC CHEMISTRY III – THEORY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

2x10 = 20

(a)	What do you understand by the relative and absolute system of configuration?
(b)	What are the conditions required for optical isomerism in an organic compound?
(c)	Define specific rotation.
(d)	What is the difference between stereospecific and stereoselective reactions?
(e)	Write down the structure and medicinal uses of thiophene.
(f)	Arrange furan, thiophene, and pyrrole in ascending order of aromaticity.
(g)	Why is oxazole less reactive than imidazole?
(h)	Differentiate between the racemic mixture and meso compounds.
(i)	Compare the basicity of pyridine with that of aliphatic amines.
(j)	Write down the synthetic applications of the Birch reaction.

SECTION B

2. Attempt any *two* of the following:

10 x 2 = 20

(a)	Define racemic modification and explain various methods for the resolution of racemic modification.
(b)	Explain in detail the nomenclature in heterocyclic compounds.
(c)	Discuss in detail the structure, synthesis, properties, and medicinal uses of pyridine.

SECTION C

3. Attempt any *five* part of the following:

7 x 5 = 35

(a)	Write a detailed note on the element of symmetry in an organic compound.
(b)	Explain conformational isomerism in cyclohexane.
(c)	Explain the following in detail. i. Fisher indole synthesis ii. Wolff-Kishner reduction
(d)	Discuss the structure and reactions of furan.
(e)	Discuss in detail the Clemmensen reduction and its synthetic applications.
(f)	Explain the following in detail. i. Birch Reduction ii. Hantzsch pyridine synthesis
(g)	Write a detail about the Oppenauer oxidation along with its synthetic applications.